银湾研究院（GreyBay Institute）
Great Bay Area Innovation and Research Center

A Scoping Review of Interventions to Improve Occupational Safety and Health of Construction Workers

EDITOR(S): *Hana Hayashi*

CONTRIBUTOR(S): *Yue Li, Shujuan Chen*

REVIEWER(S): *Wei Li*

PUBLISHED: Nov 30th, 2023

REPORT NO.: GBI-OH-2023-01

SERIES: GreyBay Occupational Health and Safety Research Series

INSTITUTION: GreyBay Institute, Shenzhen, China

KOMATSU

This study was funded by KOMATSU Ltd.. The sponsor had no control over the interpretation, writing, or publication of this work.

© 2022 GreyBay Institute

This report is licensed under Creative Commons Attribution 4.0 (CC BY 4.0).

Abstract

Background: Construction is a high-risk industry in terms of occupational safety and health issues. However, to our knowledge, there is no review to date that comprehensively examines the evidence on occupational safety and health for construction workers. This review serves a good compliment to current regulation systems and a good reference for the practice in the construction industry.

Method: This review carried out the search in 2 electronic databases, PubMed and Web of Science. Selected publications are in English and were published between January 01, 1990 and December 01, 2019.

Result: 1297 articles were retrieved and 24 were selected for final evaluation. 10 studies were randomized controlled trial (RCTs), 9 were cluster RCTs, and 5 were pre/post or longitudinal non-randomized studies. For intervention aims, 4 studies were safety-related targeting accident reduction, and 23 were health-related targeting lifestyle changes or exposure reduction. For effectiveness, 17 studies reported significant intervention effects, while 7 found their primary outcome not significantly improved.

Conclusions: This review concludes that there are a variety of effective interventions covering different aspects of occupational safety and health of construction workers. Future research should spare more effort in mental health and focus on the interaction effect of safety-related interventions and health-related interventions by implementing them together as a combination. Better bias control is needed by performing randomization and blinding thoroughly, using more non-self-report outcome data, and select randomization level carefully to reduce contamination.

Table of Contents

Introduction *5*

Methods *6*

Protocol **6**

Inclusion and exclusion criteria **7**

Information sources and search strategy **7**

Data extraction and quality assessment **8**

Risk of bias in each study **8**

Results *8*

Study selection **8**

Study characteristics **10**

General characteristics 10

Intervention method 10

Intervention aims 10

Study designs **11**

Inclusion and Exclusion criteria 11

Intervention content and duration 12

Outcome assessment 12

Result of individual studies **13**

Risk of bias within studies **13**

Randomized studies 14

Non-randomized studies 16

Discussion *16*

Study design and bias control **17**

Safety related interventions **18**

Physical health and lifestyle interventions **19**

Mental health **20**

Limitations **20**

Conclusions and recommendations **21**

Acknowledgement ..... *Error! Bookmark not defined.*

Author contribution ..... *Error! Bookmark not defined.*

Funding ..... *Error! Bookmark not defined.*

Tables ***22***

Table 1. General characteristic of included studies (n = 24) 22

Table 2. Study design of included studies (*n* = 24) 26

Table 3 Result of intervention effectiveness in included studies (n = 24) 47

Table 4 Risk of bias in included RCTs (n = 19) 67

Table 5 Risk of bias in included non-randomized studies (n = 5) 69

References ***70***

Introduction

Construction industry is considered one of the most dangerous industries due to its high injury and mortality rates. In 2005, the International Labor Organization (ILO) estimated that more than 60, 000 people die in the construction industry each year globally. [1] In developed countries, 25-40% of occupational deaths take place at construction sites, despite the fact that the construction industry employs only around 6-10% of all workers. [1]

Besides occupational safety, construction is also a high-risk industry in terms of health issues. Construction workers are more likely to result in health consequences like physical disorders, skin diseases, respiratory disorders, and cancer. Some of these conditions arise because construction workers are exposed to various chemical-related dangers, along with physical hazards such as dust, fibers, noise, radiation and extreme temperature. [2] The report of ILO states that, around 30% of construction workers are suffering from back pain or other musculoskeletal disorders. [1] Diseases like cancer, heart disease, cerebrovascular disease, and pneumonia account for 66.5% of male and 69.6% of female mortality among construction workers [3]. Among the 818 cases of deaths in a cohort consisted of 19943 male Germen construction workers that was followed for 10 years, 299 workers died of cancer, 138 died of heart diseases, whereas only 52 died of accidents. [4]

Surprisingly, though it is clear that safety and health are both crucial to the wellbeing of construction workers, deaths due to health problems occupies a much larger proportion of all-cause mortality among construction workers. This could be explained not only by their elevated exposure to hazardous sources, but also by the higher morbidity risk determined by their socioeconomic status. [5] Evidence has shown that from 1992 to 2006, the prevalence of being an ever smoker and the prevalence of being a current daily smoker are both the highest in construction workers

compare to other blue-collar workers, which is also far higher than white-collar workers. [6] Low income people also have lower quality in their diet because healthier food usually have higher price. [7]

Previous efforts of protecting the safety and health of construction workers could be seen in the regulations in countries over the world. It helped reducing the certain number of accidents and morbidities. There is existing evidence regarding health and well-being for construction workers. For example, a study by Sorensen in 2004 showed that integrating health promotion and occupational health and safety result in higher effectiveness than an occupational health and safety intervention alone. [8] The conclusion of this study was included into the report of National Institute for Occupational Safety and Health (NIOSH), serving as a guide for future practice of regulation implementation. [9]

Despite the number of intervention studies is high, however, to our knowledge there is no review to date that comprehensively summarize the result of academical intervention studies on occupational safety and health for construction workers. This review aims at summarizing the findings of present intervention studies and giving suggestions to future studies and worksite practices.

Methods

Protocol

This systematic review was conducted based on PRISMA [10] guideline, which is described below.

Inclusion and exclusion criteria

Inclusion criteria of this review include: being an intervention study that targets personnel related to construction industry; having at least one primary outcome that aims at improving occupational safety or health; published between January 01, 1990 to December 01, 2019; and written in English. Studies from any continent, country or area were all included regardless of geographical borders. For exclusion criteria, we excluded protocol studies and process evaluation studies.

Information sources and search strategy

This study was conducted through a search in electronic databases in December 2019. PubMed, Web of Science were searched with time limitation of from January 01, 1990 to December 01, 2019.

The following search strategy was applied for both PubMed and Web of Science database: (“program” or “prevention” or “intervention” or “measure” or “training” or “effect” or “implement” or “tool” or “preventing”) AND ("civil engineer" or “construction” or “green job” or “green building” or “brick layer”) AND ("trial"). For the search on PubMed, filter used was the following: Article types: Clinical trial; Randomized Controlled Trial; Species: human; Language: English. For Web of Science, filters were: Language: English; Document type: article. The search was conducted in English. The above search strategy was the final version after a pilot search, and the formula was modified with reference to those papers came out in the pilot search to ensure maximum inclusion.

Data extraction and quality assessment

Three authors independently carried out the process of reviewing the title, abstract and full text and extracted all critical data from included studies. Extracted data include author, country of the study, publication year, study design, sample size (participants or construction sites), intervention aim or topic, intervention method, study object, inclusion / exclusion criteria, experimental group (content of intervention), control group, outcomes measured, baseline and end-intervention result, interpretation of the result or conclusion of the study, strengths and limitations considered by the authors.

Risk of bias in each study

To assess the risk of bias in RCTs, the Cochrane Collaboration's tool [11] was used. Non-randomized studies were assessed for risk of bias using the RoBANS tool [12]. The criteria of Cochrane Collaboration's tool include sequence generation, allocation concealment, blinding of participants, personnel and outcome assessors, incomplete outcome data, selective outcome reporting, and other potential threats to validity. The criteria of RoBANS tool include the selection of participants, confounding variables, measurement of intervention (exposure), blinding of outcome assessment, incomplete outcome data and selective outcome reporting. A final evaluation is included for both tools as is required.

Results

Study selection

A total of 1297 citations were retrieved upon searching the selected two electronic databases.

416 records were from PubMed and 881 records were from Web of Science. 1159 records remained after removing duplicates. Then, 131 records were selected after screening by title for full-text evaluation. Finally, 24 studies [13]–[36] remained for final qualitative evaluation, all with acceptable level of quality. The PRISMA flowchart of this screening process is depicted in Figure 1.

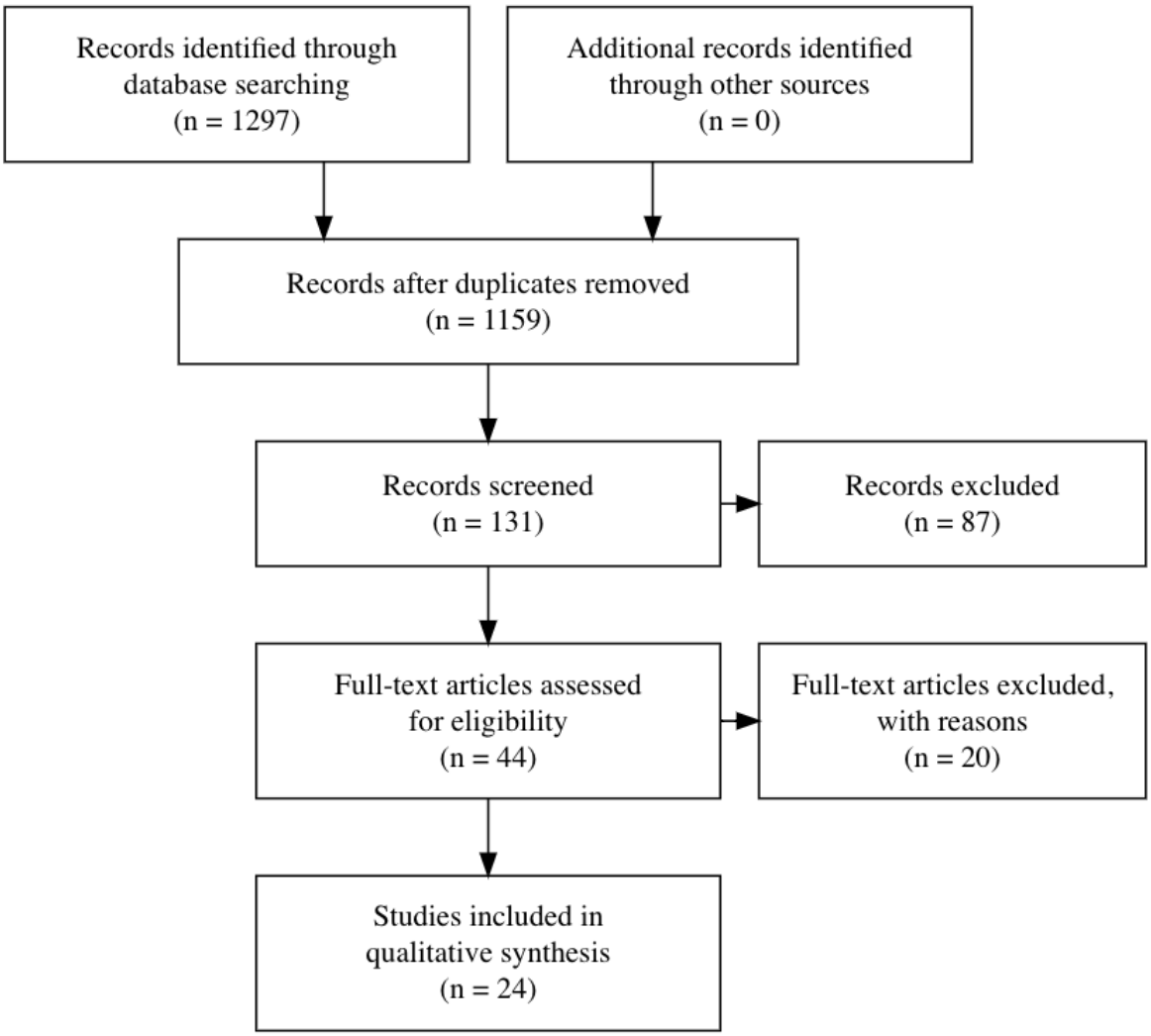

Fig 1. PRISMA flow chart of the selection process

Study characteristics

General characteristics

General characteristics for each study are summarized in Table 1. All of the 24 studies were published between 2003 and 2019, among which 20 studies were published between 2010 and 2019. In terms of the countries where the studies were conducted, 10 studies were in Netherland, 8 studies were in USA, 3 in Denmark, 1 in China, 1 in Australia and 1 in UK. Participants enrolled in the studies range from 67 to 1638, and the involved companies or sites range from 1 to 50.

Intervention method

The methods by which the interventions were implemented include personnel-dependent methods like training, counseling, education, coaching, discussion and inspection, and also non-personnel-dependent ways like video, pamphlets, text messages. Most of the studies targeted at construction workers, but a few of them targeted at managers or supervisors. Due to the fact that gender distribution is highly uneven in constuction industry, although some the studies intended to target on participants regardless of their sex, many of them eventually targeted only male workers.

Intervention aims

According to the aim and targeted outcomes of each studies, they were separated into categories of safety and health. Though safety and health are highly connected, we define safety related interventions as interventions with aims of accident reduction, while healthy related interventions target relatively long-term wellbeing including lifestyle changes and chronic disease

prevention. Based on this classification criteria, only four studies were recognized as safety related and cover topics of fall prevention, injury reduction, sun safety, safety climate, safety planning and organizational context.

23 studies were health related, and they cover a much broader range. This include lifestyle behaviors, exposure reduction like silica, quartz or noise, contraceptive use, aerobic capacity and muscle strength, physical functioning, and also mental health like stigma or social support. For lifestyle interventions specifically, 8 studies were found to have at least one outcome on lifestyle changes. The scopes that were covered include contraception, tobacco cessation, consumption of fruit and vegetables, recreational physical activity, consumption of snacks and sugar-sweetened beverages, sickness absence and work-related vitality. Mental health related interventions were only studied in three of the included publications, whose topic were sitgma, work engagement and organizatinal context (mainly social support from supervisors and the teams).

Study designs

19 studies were RCTs [13], [15], [24]–[27], [29], [31], [34]–[36], [16]–[23], among them 9 were cluster RCTs that randomized at company, site, gang or group level [13], [18], [20], [25]–[27], [29], [31], [35], and 5 were pre/post or longitudinal non-randomized studies [14], [28], [30], [32], [33].

Inclusion and Exclusion criteria

All studies listed their inclusion and exclusion criteria. Studies with cluster RCT design also listed their criteria at the cluster level. Majority of the criteria include language, age, whether giving consent, and completion of baseline or final outcome measurement. Several studies gave more specific exclusion criteria like having too many sick leaves [15], [23], or having other

unrelated serious disease and health conditions [18]. Two studies focusing on the reduction of new onset of shoulder pain excluded participants with previously occurred shoulder pain [33], [36]. Houdmont's study on sun safety excluded participants that exposed to sun light too little [30], and two studies aimed at increasing muscle strength and reducing pain excluded workers who worked less than 20 hours per week [21], [24].

Intervention content and duration

The duration of intervention could be classified to two groups, one is minute and hour levels and could be considered as a one-time intervention, while the other is usually on the scale of week, month or even year. Three studies were considered as a one-time intervention [14], [18], [30]. The duration of interventions listed in Table 2 does not include follow-up period during which no difference exists between intervention group and control group. The content of each intervention could have several sections, and it is hard to generalize across studies. Detailed contents are listed in Table 2. Contents delivered to control groups are mostly no intervention or usual care. Some of the control groups were given brief lecture, brochures, handouts, which were considered to have minimal effect compared to the intervention on the outcomes [18], [19], [24], [32]. Milner in 2018 gave control group the same material delivered to the intervention group at the end of the study [22].

Outcome assessment

Outcome measured could be classified to self-reported outcomes (e.g., safety knowledge, fruit and vegetable consumption) and outcomes with actual physical measurement (e.g., body weight, BMI, heart rate). The definition of a self-reported outcome could be considered as not measuring outcomes via devices, observation records, or diagnoses. Some studies are mixed with both self-

reported outcomes and non-self-reported outcomes. Only 8 studies whose outcomes were non-self-reported. [15], [16], [19]–[21], [23], [26], [31]. Self-reported data were collected via previously validated questionnaire or questionnaire designed by the researcher themselves.

Result of individual studies

Among the 24 included studies, 17 studies reported significant effectiveness in their tested intervention [13], [14], [30]–[36], [15]–[21], [25], and 7 found their primary outcome not significantly improved compared to control group or pre-intervention stage [22]–[24], [26]–[29]. A summary of the intervention effectiveness with data of primary outcome, interpretation of the data or conclusion of the study, author's view of strengths and limitations is presented in Table 3.

All the 7 studies with insignificant improvement in their primary outcome were occupational health related studies. Among them, 3 studied mental health outcomes and 5 were about work ability or muscle strength. All 4 studies related to occupational safety resulted in significant result. Possible reasons for ineffectiveness can be classified into program failure and theory failure. [39] Program failure refers to the poor implementation of the intervention that results in an insignificance of outcomes changes. Theory failure means that the intervention was well implemented, but the changes was not significant. We can see both program failure, such as low compliance to the intervntion, low satisfaction, low economic climate, and theory failure, such as low impact intervention design, insufficient sample size, exist across these studies.

Risk of bias within studies

To assess the risk of bias in the 19 RCTs and cluster RCTs, the Cochrane Collaboration's tool [11] was used. The results for each study were presented in Table 4. The overall risk of bias is very high in the included RCTs and cluster RCTs. Only 10 studies could be considered as low risk of

bias.

Randomized studies

All RCT studies met the criteria of adequate sequence generation and incomplete outcome data addressed. For free of selective reporting, the two studies by Viester were found to be the same intervention, but the outcomes were split into two groups. [15], [23] One only reported effective results and the other reported only ineffective results. The same situation happens to the by Oude Hengel [26], [27], and another pair of study by Gram [21], [24].

In the context of construction industry, participants can rarely be blinded in intervention studies, almost all studies failed to meet the criterion of blinding the participants to intervention allocation. Participants had to give consent in order to be enrolled in intervention studies, which means they could not be blinded for intervention that applied to them. This hold true for most of the studies except one in Milner's study [22], in which it was claimded that the participants were blinded for the intervention that based on text messages. However, as long as the intervention content need to be learned by participants, they could more or less experience a different psychological implication.

Few studies were considered successful in blinding the outcome because self-reporting was common, typically in Sorensen's [34], Groeneveld's [17], and Hammer's [29] study. The adoption of self-reported outcome measurements might introduce recall or self-report bias since the unblinded participants might have a tendency to report higher level of outcomes to meet the expectation of the researcher because of higher involvement of the intervention. Filling out questionnaires related to work ability makes participants feel sensitive is another aspect of how subjectivity affect outcome assessment happened in Oude Hengel's study. [20] Several attempts

have been made to avoid the use of subjective questionnaires. Non-self-reported measurements like observation, which is adopted in Deurssen's study, could partially reduce this bias. [31] Seixas's study adopted task card to let workers keep record of their HDP use, which is a smart way to avoid relying on recalling. [25] Brandt's study used technical measurements to quantify the workload rather than relying on self-reports or observations. [18] Viester's [15], Groeneveld's [16] and Gram's study [21] used laboratory test for objective parameters like blood pressure, blood cholesterol, and BMI.

For allocation concealment, a study was considered free of this bias only if it clearly stated that personnel were blinded. Allocation concealment is also performed less than expected with only seven studies mentioned that data analysis perosonels were blinded.

Study was rated high in risk of other bias source if significant selection bias, retention bias resulted from too many loss-to-follow-up, or failure of randomization was identified. For example, in Groeneveld's study, around half of the participants have incomplete data on body weight, absenteeism, direct health care costs, direct non-healthcare costs, and indirect non- healthcare costs, and were imputed by multiple imputation, which might significantly decrease the validity of the intervention effect. [19] Also, in especially cluster RCT, for example in Sparer's study [13], though randomization is performed, limited site number caused high difference in population. In this study, only workers on-site for >30 days are selected for analysis to reduce the effect of contamination due to worker's mobility, which may on the other hand introduce selection bias. In Seixas's study, randomization for baseline HPD use failed. [25] And in Groeneveld's study, a 25%~30% dropout rate was relatively high after randomization. [17]

Non-randomized studies

For the 5 non-randomized studies, the RoBANS tool [12] was used to assess their risk of bias. Most of the non-randomized studies were assessed as high risk of bias. The detailed assessments are presented in Table 5. Major risk of these studies was identified to be the participants selection process and absence of randomization.

In Marin's study, Participants are recruited based on their willingness, this willingness itself could be a huge selection bias source. This study also cannot avoid the bias introduced by inability of blinding participants. [14] In Houdmont's study, similar selection bias result from the recruiting process is also a concern. In the same study, contamination was overserved between groups because workers in the control group were found imitating the behaviors of participants in the intervention group, for example, finding shelter to avoid sun light exposure. [30] Dian's study compared two intervention arms, but did not set up a pure control group. [32] Borstad's longitudinal study does not randomize participants as RCT, both exposure and outcome are not blinded, and the interval of measuring outcome is too large to be used to derive a reasonable intervention effect comparing to the non-intervention group. [33] In de Boer's study, the absence of randomization lead to huge difference of participants characteristics in two groups at baseline. The adjusted disability pension receiving rate is almost significantly higher in intervention group. High loss-to-follow-up and unbalanced number of participants in two groups are also risk factors. [28]

Discussion

This review provided a comprehensive evaluation of the effects of different kinds of

intervention that aim at promoting different aspects of occupational safety and health of construction workers. Results showed that most of the intervention tested had a positive effect on the participants, and majority (17/24) of the studies were significantly effective in improving the primary outcomes of the interventions. However, more discussion could be made in terms of study designs, participants selection, outcome measurements, results interpretations and bias control.

Study design and bias control

The special context in construction industry made it very difficult to maintain the quality as the studies in other fields. The major characteristics of construction workers that differ them to participants in other studies are that (1) workers may keep transporting from sites to sites; (2) workers work together with each other. Mobilization happens more as the duration of interventions get longer, which introduces contamination in the intervention or control group or causes loss-of-follow-up. On the other hand, workers working together will influence each other's behavior which might also introduce contamination if randomization took place at individual level.

Thus, whether to choose the unit as site or individual seems to be a trade-off, and which study design reduces contamination more should be based on the method and topic of the intervention. However, realistically, selection of participants is often connected to the cooperation with the construction sites, and delivering intervention in groups reduces cost, which are some reasons why around half of the studies performed randomization at site level. This caused another problem which is the shrinkage of the number of randomization units. Among the 9 cluster RCTs, three have a site number of 8, four have 15 sites, one for 10 sites and one for 20 sites. And this is why randomization failure or heterogenity of participants characteristics at baseline between intervention group and control group are common problems in these studies.

Though little could be done in blinding particiapnts from intervention exposure, reliability of outcome assessment could be improved by choosing more objective assessing method. Objective measures are not subject to a large degree of individual interpretation, are likely to be reliably measured across patients in a study, and have higher reproduciablity and comparibility across different studies. [38] Nevertheless, even in observational studies, compliance occurs with intervention recommendations because workers know they are being observed, as is implied by Hawthorne effect [37]. Laboratory tests, for example blood pressure and blood cholesteral, are considered to have the lowest risk of bias. When outcome could not be assessed through laboratory test, validated questionnaires, for example Shoulder Pain and Disability Index (SPADI), could reduce bias from subjective assessment by participants or clinician. When all above methods are not applicable, keeping record is a way to reduce bias comparing to questionnaires based on recalling. It is highly recommended that future studies should consider using more objective method of less bias.

Safety related interventions

There were only four studies that were safety related. The low number could be explained by several reasons. First, accident prevention systems in especially developed countries are already well established. If there is any effective method to reduce accidents, it might have been incorporated into the regulations already. Under the present structure of safety regulations, it is hard both to test and to implement the intervention and figure out how much these interventions contributed to outcomes.. Second, it might be difficult to take accident numbers as outcomes in a small-scale study (usually short-term, or one time intervention) with few hundred participants. Using safety climate index or testing safety knowledge is not considered as a straightforward

method to measure physical outcome. Thus, the real effectiveness of safety related interventions is hard to evaluate.

Nevertheless, safety is the foundation of health. As long as construction workers don't feel they are safe enough, it is difficult to promote their health through behavioral change. In Sorensen's study, it was discussed that persuading workers to quit smoking was difficult when they believe their lives are constantly in danger, which made them think it was preferable to trade future health for present joy of smoking. [34] Thus, more safety related intervention studies in the future are need, targeting what construction workers are worried about, what kind of cases contribute to the of accident statistics, and what are implementable and incorporable to current safety regulations.

Physical health and lifestyle interventions

Lifestyle interventions with ultimate goal of physical health promotion is a main component of health-related interventions. There are already established evidence that construction workers and/or blue-collar workers tend to have disadvantage in their health. It has been shown that blue-collar workers are more likely to be smokers than white collar workers [6], and also blue-collar workers found it harder to quit smoking [40]. Additionally, people with less education and low incomes are less likely to show adherence to the Mediterranean diet [41]. Construction worker population, with a majority of male, are also less likely to participate in worksite health promotion programmes. [42] However, occupational safety and health are profoundly interconnected, and so do physical health and lifestyle interventions. As stated in previous section, safety insurance serves as the foundation of health promotion, and similarly, basic physical health protection is the basis of adopting lifestyle changes. The mechanism of lifestyle interventions is by arousing worker's subjective motivation of behavioral changes. Thus, it might be more effective and efficient to

implement lifestyle interventions with resolving safety issues as a prerequisite.

Occupational safety, occupational health, and healthy lifestyle are thus arranged in an order of more long-term consideration of worker's well-being, and a higher hierarchy of ambition in terms of promoting construction worker's safety and health. It is suggested that future lifestyle interventions could include safety interventions and examine possible interaction effect between lifestyle interventions and safety-related interventions on safety and health outcomes.

Mental health

There is a high need in improving construction worker's mental well-being. It is estimated in a study that the prevalence of self-reported mental health effects among bricklayers and supervisors, respectively, were as follows: high need for recovery after work (14%; 25%), distress (5%, 7%), depression (18%, 20%) and post-traumatic stress disorder (11%, 7%) [43]. In the same study, high work speed and quantity were found to be associated with symptoms of depression. Unfortunately, there were only three studies with ineffective results in this review related to mental health. The insufficient evidence of effective intervention targeting mental health and the high needs for these interventions calls for further research in this area.

Limitations

There are several limitations of this review. One of the limitations is that the database used for searching is limited to PubMed and Web of Science only. We did see an overlap of the records at first level of screening, but the overlap is rather a small portion, indicating that some other studies that satisfy the criteria might be overlooked. Another limitation is the language of the search was limited to English, so there might have been other articles that were not included. Third, due to the variety of topics and aims that the interventions covered, it is not possible to perform a

quantitative analysis. Nevertheless, this review could serve as a good reference for future reference of intervention studies in this field, and also for recommendation to current practice of construction industry.

Conclusions and recommendations

This review concludes that there are a variety of effective interventions covering different aspects of occupational safety and health of construction workers. Future research should spare more effort in mental health and focus on the interaction effect of safety-related interventions and health-related interventions by implementing them together as a combination. Better bias control is needed by performing randomization and blinding thoroughly, using more non-self-report outcome data, and select randomization level carefully to reduce contamination.

Tables

Table 1. General characteristic of included studies (n = 24)

Study	Publication year	Country	Participant number / Site number	Intervention Aims / Topics	Intervention Method	Intervention target
Sparer [13]	2016	USA	615 / 8 sites	Improvement of safety climate	Inspection, communication and recognition	Construction workers and construction site
Marín [14]	2017	USA	118	Fall prevention, silica exposure, leadership, communication, and safety planning	Training on knowledge and skills	Construction supervisors
Seixas [25]	2011	USA	176 / 8 sites	Promotion on the use of hearing protection devices (HPDs)	Training on knowledge, use of device and toolbox	Construction workers
Houdmont [30]	2016	UK	120	Sun safety	Video	Construction workers
Deurssen [31]	2015	Netherlands	282 / 8 companies	Quartz exposure	Presentation, documentary and videos, skills training	Managers and construction workers
Dian [32]	2012	China	1638	Promoting contraceptive use	Pamphlets, posters, free condom distribution, lectures, counseling,	Unmarried male construction migrants

					videos, peer education, hotlines	
Borstad [33]	2009	USA	240	Reduction of shoulder pain	Stretching and strengthening exercise	Construction apprentices
Sorensen [34]	2007	USA, Canada	582	Promotion of smoking cessation and increasing fruit and vegetable consumption	Telephone counseling, mailed report, educational materials	Construction workers
Peters [35]	2018	USA	607 / 10 sites	Improvement on ergonomics practices, reduction in new pain and injury, and improvement on diet and physical activity	Inspections and feedback, task-preplanning, training, education and coaching	Construction workers
Ludewig [36]	2003	USA	67	Reduction of pain and improvement of shoulder function	Home exercise programme	Male Construction workers
Viester [15]	2018	Netherlands	267	Improvement on lifestyle behavior, health indicators, and prevention and reduction of overweight	Individual coaching sessions, tailored information, and materials	Male Construction workers
Groeneveld [16]	2010	Netherlands	517	Improving energy balance-related behavior or smoking cessation (non-self-reported)	Individual counseling	Male Construction workers
Groeneveld [17]	2011	Netherlands	595	Improving energy balance-related behavior or smoking cessation (self-reported)	Individual counseling	Male Construction workers

Brandt [18]	2018	Denmark	80 / 15 gang	Physical workload	Workshops (mainly discussions)	Male Construction workers
Groeneveld [19]	2011	Netherlands	573	Lifestyle intervention for workers at risk for cardiovascular disease	Individual counseling	Male Construction workers
Oude Hengel [20]	2014	Netherlands	293 / 15 departments	Health and work ability (reduction of physical workload)	Individual and worksite training on skills and awareness	Construction workers
Gram [21]	2012	Denmark	67	Aerobic capacity and muscle strength	Individualized worksite exercise training	Male Construction workers
Milner [22]	2018	Australia	682	Mental health stigma	Text messages with embedded hyperlinks via smart phone	Male Construction workers
Viester [23]	2015	Netherlands	314	Musculoskeletal symptoms, physical functioning, work ability, work-related vitality, work performance, and sickness absence	Tailored lifestyle information and coaching, exercise instructions, and toolbox	Construction workers
Gram [24]	2012	Denmark	67	Aerobic capacity and muscle strength	Individualized worksite exercise training	Male Construction workers
Oude Hengel [26]	2013	Netherlands	293 / 15 sites	Work ability, health, and sick leave	Individual and interactive training	Construction workers
Oude Hengel [27]	2012	Netherlands	293 / 15 sites	Work engagement, social support, physical workload, and need for recovery	Individual and interactive training	Construction workers

de Boer [28]	2007	Netherlands	292	Work ability and work disability pension	Counselling and education	Construction workers
Hammer [29]	2019	USA	292 / 20 groups	Organizational context	Supervisor training, behavior tracking and discussion	Supervisors and construction workers

Table 2. Study design of included studies (*n* = 24)

Study	Design/sample size (n) / retention rate (%)	Inclusion/exclusion criteria	Experimental group		Control group	Outcomes measured
			Intervention duration	Contents	Contents	
Sparer [13]	Cluster RCT/Total: 615 IG(n=494-433), CG(n=211-182)/87%	Sites: Sites had to be using the online data inspection management program Predictive Solutions as standard practice prior to study initiation. A site had to operate for >4 months from study initiation and have ≥30 workers at any one time. Workers Aged 18-65 who could read and write English. Only those workers completed both pre- and post-exposure surveys are included.	4-6 months	B-SAFE program, primary components are: (i) weekly worksite safety inspections; (ii) weekly feedback and communication; (iii) monthly recognition and reward.	Contractor's standard safety programs along with a few posters with the program logo only.	The mean safety climate score at intervention sites, by surveying workers at the sites
Marín [14]	Pre-post/n=118	All foremen, supervisors, contractors, and workers from more than 50 construction firms across the Greater Lawrence Area	5 hours	5 hours training of Leaders in Safe Construction (LISC): Trained construction	NA	Safety-efficacy: safety knowledge, attitudes, and

		that attended the construction supervisor safety training were included in the study. Exclusion criteria before intervention began were not stated.		supervisors in safety leadership and planning, fall prevention techniques and requirements, and communication strategies.		skills, by surveying participants
Seixas [25]	Pre-post (for baseline training) AND Cluster RCT (for toolbox)AND RCT (for NLI)/Total: (sites=8, n=176) No Noise indicator 4 sites: baseline (n=77-46), baseline & toolbox (n=78-44) Noise indicator 4 sites: baseline (n=57-45), baseline & toolbox (n=59-41)	Sites: Large commercial construction sites with unionized labor, that ensure participant numbers and intervention duration and assessment period. Workers: About 30 subjects were recruited at each site. Only subjects who returned for baseline training and completed all three assessments are included in the study sample.	2 months	(1) Baseline training on hearing loss prevention and HPD use. (BL) (2) With or without follow-up 'toolbox' reinforcement training. (TB) (3) With or without personal noise level indicator. (NLI)	One-time baseline training	Percent of time HPD used, workers using HPD 50% of the time.

Houdmont [30]	Cross sectional intervention study/Total n=1279-160-120 IG (n=70), CG (n=50)/18%	Workers who provided contact details, returned a completed baseline and follow-up questionnaire. Excluded if no outdoor work being reported or no information given on intervention exposure.	12 minutes	A 12-min DVD titled Sun Safety in Construction: A Workplace Health Guidance Film.	Workers who completed baseline questionnaire and follow-up questionnaire but did not receive the intervention.	Correct sun safety knowledge, Percent in taking sun safety actions on a typical summer workday
Deurssen [31]	Cluster RCT/Total: site 13-8, workers N=116, measurement n=149 IG (N=63-55, n=81-70), CG (N=53-49, n=68-63)	Sites: 13 companies were approached and 5 was excluded for too few permanent construction workers or insufficient work supply during period of baseline survey. Workers: Job categories were selected based on exposure levels which include bricklayer, carpenter, concrete	6 months	(1) Plenary session: presentation, documentary about health risks of quartz exposure and PIMEX videos. (2) Individual session: teach construction workers how to use technical control measures at worksite.	Sites that were randomized to control, with no intervention implemented.	Mean exposure to quartz, mean duration of use of control measure as fraction of the time performing abrasive tasks.

		driller, demolisher, and tuck pointer.		(3) Meeting: give managers more insight in the availability of state-of-the-art technical control measures.		
Dian [32]	Pre-post/ Group A (7 sites, n=415-431) Group B (7 sites, n=388-404)	Sites: Workers employed were mainly young male migrants; the project at the sites could not be finished in 3 months from the beginning of the study; adequate number of participants (≥160) and had a relatively steady population; workers rarely communicate with each other. Workers: Unmarried male construction migrants who agreed to join the study	3 months	YOLAMI service packages. Package A: pamphlets, posters, free condom distribution. Package B: lectures, face-to-face counseling, videos, peer education, hotlines.	NA	(1) Knowledge- and attitude-related contraception between baseline and end of the study (2) Contraception-related behaviors among sexually active workers in the past 3 months (reporting using condom usually, ever had an

						unwanted pregnancy)
Borstad [33]	Longitudinal/Total: 298-240-208, IG (n=145-117-102), CG (n=153-123-106)	Workers: First- and second-year apprentices from local sheet metal, electrical and plumbing and pipe fitting trades were recruited; subjects reported shoulder pain at baseline and those who were examined and diagnosed with subacromial impingement syndrome were excluded.	Every other day for 2 years	A stretching and strengthening exercise program that includes: stretch the posterior shoulder region and the upper trapezius and pectoralis muscles five times daily, holding each stretch for 30 s; a push up plus exercise to strengthen serratus anterior and to perform resisted humeral external rotation to strengthen	No intervention	New-onset shoulder pain at year 1 and year 2.

				infraspinatus and teres minor muscles with elastic bands.		
Sorensen [34]	RCT/Total: 674-582, IG (n=281), CG (n=300)/86%	Participants were members of LIUNA and were construction laborers that completed the background survey, consented to participate in the RCT, completed the baseline survey and the final survey.	6 months	(1) one-to-one motivational interviewing counseling sessions, delivered by telephone with a health advisor, (2) a mailed tailored feedback report, and (3) written educational materials, targeted to the specific needs and work experiences of construction laborers.	No intervention	Smoking cessation and increases in fruit and vegetable consumption.

Peters [35]	Cluster RCT/Total: sites10, n=607 IG (5 sites, n=324) CG (5sites, n=283)	Sites: Worksites had to be in operation for 4 months or longer and have 30 or more workers. Workers: Aged 18–65, and were English literate, self-nominated and provided verbal consent during the survey process. All subjects gave their informed consent for inclusion before they participated in any data collection activities.	6 months	(1) Soft Tissue Injury Prevention Program (StIPP) which focused on improving ergonomics practices at the site and worker level to improve musculoskeletal health; and (2) Health Week, that integrated key messages and provided integrated health coaching opportunities for individual workers to improve ergonomic practices and improved health behaviors (diet, physical activity, and smoking) associated with cardiovascular health.	No intervention	Pain and injury, dietary and physical activity behaviors, smoking, ergonomic practices, and work limitations

Ludewig [36]	RCT/Total: 67, Symptomatic subjects: IG (n=34-30) CG (n=33-32) Asymptomatic subjects: (n = 25-23)	Inclusion: (1) All subjects were required to have a self-reported occupational exposure to overhead work for longer than one year and a minimum of 130˚ of active scapular plane abduction as measured goniometrically during a clinical examination. (2) For symptomatic individuals: A current reported history of shoulder pain localized to the glenohumeral joint region excluding cervical and periscapular pain, but including the common site of referred pain of the rotator cuff to the C5–6 dermatome above the deltoid insertion. Present with at least two positive shoulder impingement tests and pain reproduction during two of three additional categories of clinical tests. Exclusion:	8 weeks	Home exercise programme. A, posterior shoulder stretch; B, pectoralis stretch; C, upper trapezius relaxation exercise, avoiding a shoulder shrug while raising the arm; D, serratus anterior strengthening exercise; E, external rotation strengthening exercise, left is arm at the side; F, external rotation strengthening exercise progression with arm abducted.	Usual care	Shoulder Rating Questionnaire (SRQ) score and shoulder satisfaction score, Shoulder Pain and Disability Index (SPADI)

		(1) had a history of rotator cuff surgery; (2) reported a history of glenohumeral dislocation, or other traumatic injury to the shoulder; (3) reported only periscapular or cervical pain during arm elevation; or (4) had shoulder symptoms reproduced by a cervical assessment.				
Viester [15]	RCT/Total: 314, IG (n=162-139-130-128) CG (n=152-138-130-129)	Inclusion: Consenting blue-collar workers of a construction company who attended the PHS. Exclusion: being on sick leave for >4 weeks at baseline.	6 months	The "VIP in construction" program, a tailored program to the noncompulsory periodic health screening (PHS) outcome, including	Received care (PHS only) as usual and was only contacted for the	Body weight, body mass index (BMI), waist circumference, physical activity (PA) levels,

				personal health coaching, information, and tools to support changes in PA and dietary behavior.	baseline and follow-up measurements.	dietary behavior, blood pressure, and blood cholesterol
Groeneveld [16]	RCT/Total: 816-517, IG (n=408-261) CG (n=408-256)	Inclusion: Male workers in the construction industry aged 18–65 with an elevated risk of CVD (a higher than moderate 10-year risk of coronary heart disease based on the Framingham risk score, and having one or more additional risk factors, i.e. body mass index (BMI)≥30; HbA1c≥6.5%; not meeting the physical activity guidelines; heart complaints; psychological complaints; alcohol intake ≥35 glasses per week.)	6 months	(1) After having discussed the CVD risk profile and current health status, the participant chose to discuss diet and/or physical activity, or smoking cessation. (2) Pros and cons of behavior change, and willingness, readiness, and perceived confidence in the ability to change were discussed. (3) The participant set long- and short-term goals, and formulated	Usual care, consisting of brief oral or written information from the occupational physician about their CVD risk profile.	body weight, BMI, systolic and diastolic blood pressure, HDL cholesterol, total cholesterol/HDL cholesterol ratio, and HbA1c

				implementation intentions.		
Groeneveld [17]	RCT/Total: 816-595, IG (n=408-288, energy balance-related behaviors (EB): n=208, smoking cessation (SC): n=80) CG (n=408-307, EB: n=225, SC: n=82)	Inclusion: Male workers in the construction industry aged 18–65 with an elevated risk of CVD (a higher than moderate 10-year risk of coronary heart disease based on the Framingham risk score, and having one or more additional risk factors, i.e. body mass index (BMI)≥30; HbA1c≥6.5%; not meeting the physical activity guidelines; heart complaints; psychological complaints; alcohol intake ≥35 glasses per week.)	6 months	(1) After having discussed the CVD risk profile and current health status, the participant chose to discuss diet and/or physical activity, or smoking cessation. (2) Pros and cons of behavior change, and willingness, readiness, and perceived confidence in the ability to change were discussed. (3) The participant set	Usual care, consisting of brief oral or written information from the occupational physician about their CVD risk profile.	PA (leisure time physical activity, sports activities, energy expenditure), diet (snacks, fruit, vegetables, alcohol), and smoking

				long- and short-term goals, and formulated implementation intentions.		
Brandt [18]	Cluster RCT/Total: gang 15, n=80 IG (7 gang, n=32) CG (8 gang, n=48)	Inclusion: Full-time construction work. Exclusion: Life-threatening diseases and hypertension >160/100 mmHg	Not mentioned (a very short one-time intervention)	A participatory ergonomics intervention with 3 workshops aimed to create possibilities for change by enabling engagement between the technical measurements and the participant.	Received handouts about WMSDs and lifting guidelines from The Danish Working Environment Authority	Situations (events) where the construction workers were exposed to excessive physical workload; physical, psychosocial, and organizational conditions; pain intensity in the last week

Groeneveld [19]	RCT/Total: 816-573, IG (n=145-293) CG (n=143-280) (imputation)	Inclusion: Aged 18 to 65 with an elevated risk of CVD, based on a screening instrument adapted from the Framingham risk score	6 months	Three 45 to 60-minute face-to- face counseling sessions at the nearest occupational health service (OHS), and four 15 to 30-minute telephone counseling sessions with an occupational physician or occupational nurse. Sessions covered participants CVD profiles, willingness and goal settings, followed by progress and barriers discussion and goal adjustment in subsequent sessions.	Received brochures on healthy lifestyle and CVD	Body weight at baseline, 6 months and 12 months.

Oude Hengel [20]	Cluster RCT/Total: department=15, n=293 IG (department=8, n=171) CG (department=7, n=122)	Inclusion: (1) Construction workers were able to complete questionnaires written in the Dutch language, and (2) construction workers provided written informed consent.	6 months	Two individual training sessions with a physical therapist to reduce physical workload, an instrument to raise awareness of the importance of rest breaks to reduce fatigue (Rest-Break tool), and two empowerment training sessions to improve the range of influence at the worksite.	Usual practice, meaning that they solely received the training sessions that Dutch companies have to organize for their employees in order to receive a Health and Safety certificate.	Primary: work ability, physical and mental health. Secondary: prevalence of musculoskeletal symptoms in different body regions.
Gram [21]	RCT/Total: 67, IG (n=35) CG (n=32)	Study participants were employees working in the construction industry, recruited from three workplaces and companies in Denmark.	12 weeks	1 hour a week (3 × 20 minutes), each training sessions included 10-minute dynamic exercises for warm-up	One-hour lecture on general health promotion.	Primary: VO2max and isometric muscle strength. Secondary:

		Inclusion: (1) The participants should have physically demanding tasks with high peak loads at work. Exclusion: (1) Working <20 hours per week.		and aerobic capacity (increasing from ~50–70% estimated maximal workload) followed by 10 minutes of individually tailored exercises.		body mass index (BMI), fat percent, blood pressure, and blood lipid profile.
Milner [22]	RCT/Total: 682-645-478, IG (n=343-247), CG (n=302-231)	Inclusion: (1) Males in the construction industry accessing services from Incolink between 30 May 2016 and 4 April 2017. (2) Owned a smartphone with internet connectivity and adequate data download capacity. Exclusion: Inadequate English language skills (a primary language other than English) and those who were under 18 years old.	6 weeks (one message / week)	The program was delivered to participants' smart phones via rich text messages with embedded hyperlinks to microsites and other digital resources such as videos and digital wallet cards. These additional resources provided information on stigma, mental health, information on help seeking, and links to sources of help, whilst also	Provided the interventio n materials in full at the conclusion of the project.	Primary: three subscales of The Self-Stigma of Depression Scale (SSDS) Secondary: suicidal ideation, communication about suicide, and suicide attempts SBQ-R.

				encouraging the establishment and maintenance of long-term contact with others.		
Viester [23]	CT/Total: 314-257, IG (n=162-128), CG (n=152-129)	Inclusion: Consenting clue collar employees of a construction company. All employees who attended a non-compulsory periodic health screening (PHS). Exclusion: On sick leave for more than 4 weeks prior to the PHS.	6 months	The programme consisted of tailored lifestyle information, lifestyle coaching sessions, exercise instructions, and the 'VIP in construction toolbox'. This toolbox consisted of an overview of the company's health promoting facilities, a waist circumference measuring tape, a pedometer, a BMI card, a calorie guide, healthy recipes, and a	Received care as usual and was only contacted for the baseline and follow-up measurements.	Musculoskeletal symptoms (Dutch musculoskeletal questionnaire), physical functioning (a subscale of the RAND-36), and work-related outcomes (work ability, work performance, work-related vitality, and sickness absence).

				lifestyle knowledge test.		
Gram [24]	RCT/Total: 67, IG (n=35) CG (n=32)	Study participants were employees working in the construction industry, recruited from three workplaces and companies in Denmark. Inclusion: (1) The participants should have physically demanding tasks with high peak loads at work. Exclusion: (1) Working <20 hours per week.	12 weeks	1 hour a week (3 × 20 minutes), each training sessions included 10-minute dynamic exercises for warm-up and aerobic capacity (increasing from ~50–70% estimated maximal workload) followed by 10 minutes of individually tailored exercises.	One-hour lecture on general health promotion.	Musculoskeletal pain in neck–shoulder, lower back, and hip–knee; work ability; productivity; perceived exertion; and sick leave.
Oude Hengel [26]	Cluster RCT/Total: site 15, n=293 IG (site=8, n=171) CG (site=7, n=122)	Inclusion: (i) construction workers were able to complete questionnaires written in the Dutch language, and (ii)	6 months	A physical and a mental component. Regarding the physical component, the	No interventio n	Primary outcomes: work ability, and mental and

		construction workers had signed a written informed consent.		workers received two individual training sessions of a physical therapist and a Rest-Break tool. As to the mental component, workers received two interactive empowerment training sessions to improve their influence at the worksite.		physical health status. Secondary outcomes: the occurrence of musculoskeletal symptoms and sick leave).
Oude Hengel [27]	Cluster RCT/Total: site 15, n=293 IG (site=8, n=171) CG (site=7, n=122)	Inclusion: (i) construction workers were able to complete questionnaires written in the Dutch language, and (ii) construction workers had signed a written informed consent.	6 months	A physical and a mental component. Regarding the physical component, the workers received two individual training sessions of a physical therapist and a Rest-Break tool. As to the mental component, workers received two interactive	No intervention	Social support at work, work engagement, physical workload, and need for recovery.

				empowerment training sessions to improve their influence at the worksite.		
de Boer [28]	Pre-post/n=292 IG (n=83) CG (n=209)	Inclusion: Intervention group: Builders aged 18–64 years and working for a company in the province of North Holland. Control group: Builders aged 40–64 years working for a company elsewhere in the Netherlands.	6 months	The programme comprised an assessment interview, additional assessment of labor market capabilities, devising an individual education programme, executing the individual education programme and a follow-up programme. The individual counselling and education programme could involve educational courses, individual counselling and	usual care	Work ability and disability pension

				coaching at the workplace.		
Hamer [29]	Cluster RCT/Total: groups 20, n=292-249-195 IG (groups=11, n=?-148-125) CG (groups=9, n=?-101-70)	Inclusion: Groups: Workgroups from a municipal public works department who were primarily field construction workers.	90 days	(1) computer-based training to teach supervisors ways to better support worker safety and work-life challenges; (2) supervisor behavioral self-monitoring to facilitate transfer of training to practice; and (3) team-based discussions with supervisors and work crew members to identify challenges and opportunities for	NA	Family Supportive Supervisor Behaviors (FSSB), team effectiveness processes, work-life effectiveness indicators, leader-member exchange (LMX), team cohesion

		improvement with 30, 60, and 90 day follow-up check-in meetings.

Table 3 Result of intervention effectiveness in included studies (n = 24)

Study	Intervention Aim / Topic	Intervention effect	Interpretation of the result and conclusion	Author's view on strengths and limitations
Sparer [13]	Improvement of safety climate	Pre-exposure IG: 42.0 (7.0) CG: 43.3 (6.4) Post-exposure IG: 42.5 (7.1) CG: 42.4 (6.2) Intervention effect +1.64 (3.28%) (p=0.01)	There is a significant intervention effect of increase in safety climate after adjustment of confounders. The intervention proves to be successful in improving safety climate, awareness, teambuilding, and communication.	Limitations: Limited site number causing high difference in population, movement of workers between study sites (low contamination), only workers on-site for >30 days are selected for analysis. The conclusions are limited to commercial construction sites with pre-existing safety programs.

Marín [14]	Fall prevention, silica exposure, leadership, communication, and safety planning	Safety knowledge: Fall prevention +0.11 (p=0.050) Silica exposure +0.28 (p=0.001) Leadership and communication +0.21 (p=0.001) Safety planning 0.00 (p=0.958) Safety attitudes: FP +0.07 (p=0.044) LaC +0.13 (p=0.021) SP +0.20 (p=0.764) Safety skills: FP +0.22 (p=0.002) LaC +0.25 (p=0.001) SP +0.17 (p=0.001)	Most dimensions showed statistically significant increase in participants' safety knowledge, attitudes and skills after training. This suggests that the LISC supervisor training was effective in improving participants' safety learning, and especially their safety leadership skills.	(1) Limitations: Intervention design, participants selection, information classification. Internal validity threats exist in pre-post survey evaluation design. Only 30% of the participation were reached for the 6-month follow-up survey. The lack of direct field observations may introduce social desirability bias that would overestimate the safety-efficacy effect. (2) Strengths: The focus on modifying the root causes of safety climate through a construction supervisor's safety leadership training is a unique contribution to the field of safety research. Participation of a large and diverse group of small construction contractors. Contact hours in training and its quality as measurement of intervention dose.
Seixas [25]	Promotion on the use of hearing protection devices (HPDs)	% of time HPD used at baseline, 2 and 4-months follow-up: All groups (34.5, 46.7, 42.0) BL Only (30.0, 43.3, 35.5) BL and toolbox (48.5, 50.3, 41.2) BL and NLI (33.7, 43.8, 42.4)	The increase in HPD use was greatest among the group receiving both TB and NLI interventions. This study demonstrates the mild impact of a well-	(1) Limitations: Failure of randomization of pre-intervention HPD use among groups, which result in great difference in initial HPD use. Accurate measurement of HPD use is difficult. Only HPD use on the day it is used was assessed rather than worker's usual practice.

		BL, toolbox, NLI (25.5, 49.6, 49.8)	constructed HPD use training and provides support for the additional use of a personal NLI to increase use of HPDs among construction workers.	(2) Strengths: task card instead of questionnaire was used, which avoid overstated use of HPD use.
Houdmont [30]	Sun safety	Insignificant for change in correct knowledge on sun safety at pre- and post-intervention. 9 out of 10 behavioral measures has significant positive change in intervention group.	Exposure to an educational intervention was linked to positive change in construction workers' self- reported sun safety practices at 12-month follow-up.	Limitations: (1) Unclear response rate and low retention rate due to nature of construction industry. (2) Baseline data were collected in a year with wettest record, yearly follow-up in future study is needed. (3) Misclassification of intervention group or control group may exist due to incorrect recall. (4) Behavior of control group might be influenced by intervention group. (5) The intervention was of a one-size-fits-all type rather than stage-matched and tailored to the needs of participants in particular stages of change.
Deurssen [31]	Quartz exposure	Quartz exposure reduction: IG: 73% CG: 40%	This study demonstrates a substantial reduction in	Limitations: (1) Workers could not be blinded both in intervention group and control group since they knew they were being observed.

		p<0.001 Percent of time control measure is used: Only significant in concrete driller IG: 62%-93% CG: 64%-69%	quartz exposure among high exposure job categories (i.e. concrete drillers, demolishers and tuck pointers) in the construction industry.	(2) Same time with the study, the Dutch labor inspectorate started specific surveys with the intention to reduce exposure, which were a likely explanation for the overall reduction in exposure that was observed in the control group also. (3) Interpretation of the result and controlling for underlying variables are difficult due to continuously changing and complex context of the construction industry.
Dian [32]	Promoting contraceptive use	Knowledge score (median): Group A: 43.8-56.3 Group B: 50.0-62.5 Attitude score: Group A: 50.0-50.0 Group B: 50.0-62.5 Used condom usually: Group A: 42/83(50.6%)-58/88(65.9%) Group B: 27/63(19.0%)-75/85(88.2%) Unwanted pregnancy: Group A: 14/83(16.9%)-13/88(14.8%) (changes not significant)	Unmarried male migrants who received the comprehensive intervention (package B) were more willing to use condoms and avoid unwanted pregnancies effectively.	Limitations: (1) Unknown factors influencing the effect of interventions may exist. (2) Recruited sexually active workers were slightly less than the calculated samples. (3) Intervention period was short, longer observation on sustainability is needed.

		Group B: 12/63(19.0%)- 4/85(4.7%)		
Borstad [33]	Reduction of shoulder pain	Overall rate for new-onset shoulder pain in the cohort of 208 subjects was 14.4%. IG: 11 (10.8%) CG: 19 (17.9%) RR=1.54 (compared to IG)	The proportion of new-onset shoulder pain in the control group was higher than in the exercise group.	Limitations: (1) Sample size, subject adherence to the exercise program, the distribution of trades by group, and the healthy worker effect might explain the minimal exercise effect. (2) Leisure time activities of the workers, psychosocial factors were not assessed; data collection occurred every 12 months, possibly resulting in subject recall bias and shoulder pain might begin but resolved during that period.

Sorensen [34]	Promotion of smoking cessation and increasing fruit and vegetable consumption	Tobacco cessation (smokers n=188): IG: 19% CG: 8% Consumption of fruits and vegetables: IG(n=280): +1.52 servings CG(n=198): -0.09 servings p<0.0001	A tailored intervention can be efficacious in promoting tobacco use cessation and increased fruit and vegetable consumption among construction laborers, a high-risk, mobile workforce.	Limitations: (1) Primary outcomes relied on self-report, and accordingly are subject to sources of reporting bias. (2) Potential of differential reporting of health behaviors between the intervention and control conditions at the final survey, given the greater involvement of intervention participants. (3) This measure does not capture smokers who quit and relapsed prior to the survey and does not capture long-term sustained cessation. (4) 5-A-Day seven-item screener, the instrument that was used to measure fruit and vegetable consumption has been added portion sizes. Strengths: (1) Extensive formative research and in close collaboration with the LHSFNA, and accordingly reflects the day-to-day realities of workers' lives. (2) Address several health behaviors simultaneously. (3) High dose delivered, high response rate and little loss to follow-up.

Peters [35]	Improvement on ergonomics practices, reduction in new pain and injury, and improvement on diet and physical activity	1-month follow-up: Ergonomics practices: B=0.20, p=0.002 Incidences of pain and injury: OR=0.58, p=0.012 6-month follow-up: Physical demanding work: B=-0.25, p=0.008 Recreational physical activity: B=35.2, p=0.026 Consumption of fruits and vegetables : B=0.87, p=0.008	The ARM program had a positive impact at the individual level on the worksites that implemented the program. The trial saw improved ergonomics practices, as well as, reduction in new pain and injury, and improved diet and physical activity, as reported by the workers.	Limitations: (1) Process evaluation revealed barriers to intervention implementation fidelity and uptake, including a fissured multiemployer worksite, the itinerant nature of workers, competing production pressures, management support, and inclement weather. (2) Intervention depended on the participation of the general contractor safety managers, whose involvement and dedication to the study varied across sites and between general contractors. (3) This study involved worksites in commercial construction only. Strengths: Cluster RCT study design and the wide variety of general contractors and sites recruited into the study.

Ludewig [36]	Reduction of pain and improvement of shoulder function	SRQ score (pretest, post-test): IG: 65.9-78.0 Symptomatic CG: 72.5-71.1 Asymptomatic CG: 93.8-94.0 Satisfaction score: IG: 4.5-6.2 SCG: 5.0-5.0 ACG: 9.1-8.8 pain: IG: 4.8-2.8 SCG: 4.6-4.1 ACG: 1.3-1.4 disability: IG: 4.1-2.5 SCG: 3.8-3.7 ACG: 1.3-1.3	The intervention group showed significantly greater improvements in the Shoulder Rating Questionnaire (SRQ) score and shoulder satisfaction score than the control groups. Average post-test SRQ scores for the exercise group remained below levels for asymptomatic workers. Intervention subjects also reported significantly greater reductions in pain and disability than controls.	Limitation: (1) The volunteer convenience sample tested may impact the generalizability of results. Subjects included male construction workers only. (2) Different from chronic intermittent shoulder symptoms, more acute subjects may respond differently to a similar exercise program. (3) Similar results should not be expected in the treatment of non-specific shoulder pain, and persons undertaking an exercise program are advised to consult a health care professional to ensure their appropriateness for an exercise intervention. (4) Intervention group received more attention of a therapist than the control group, and this differential attention may have impacted the results. (5) The apparent effectiveness of the exercises might appear to be less for subjects not seen for follow-up until the 12th week.

Viester [15]	Improvement on lifestyle behavior, health indicators, and prevention and reduction of overweight	6-month follow-up: Body weight: B=-1.06, p=0.010 BMI: B=-0.32, p=0.010 Waist circumference: B=-1.38, p=0.32 Physical activity: B=2.06, p=0.32 Sugar-sweetened beverages: B=-2.82, p=0.003 Blood pressure and blood cholesterol: insignificant	Intervention participants showed positive changes in vigorous PA and intake of sugar-sweetened beverages compared to controls, as well as effects on weight-related outcomes at 6 months. Long-term effects were still promising but not statistically significant.	Limitations: (1) Diet and PA were measured by self-report. (2) Validated food frequency questionnaires were not used. Strengths: (1) RCT study design. (2) Intervention was tailored to the individual worker.
Groeneveld [16]	Improving energy balance-related behavior or smoking cessation (non-self-reported)	6-month follow-up: Body weight: -1.9 (-2.6, -1.2) BMI: -0.6 (-0.8, -0.3) DBP: -1.7 (-3.3, -0.1) 12-month follow-up: Body weight: -1.8 (-2.6, -1.1) BMI: -0.6 (-0.8, -0.3) insignificant in other outcomes	Individual-based counseling resulted in significant beneficial long-term effects on body weight. This is an important finding for occupational health, considering the rising prevalence of obesity and CVD.	Limitations: (1) Central body fat is not measured, which is more strongly related to CVD than body weight. (2) The dropouts were on average 1.3years younger than the participants used for data analyses. (3) Counselors used specific MI techniques in the intervention, but they were not fully proficient in MI as a whole, as concluded from our evaluation of the practical execution of the intervention. Strengths: (1) Large sample size. (2)

				Randomization performed at individual level. (3) Chances of cross-contamination between groups were minimal, because they were recruited from different companies across the country. (4) The contrast between groups was large, since the compliance of participants in the intervention group was high. (5) The study can be considered as an effectiveness study as opposed to an efficacy study.
Groeneveld [17]	Improving energy balance-related behavior or smoking cessation (self-reported)	6-month follow-up: Snacks: -1.9 (-3.7, -0.02) Fruit: 1.7 (0.6, 2.9) 12-month follow-up: Snacks: -1.9 (-3.6, -0.2) insignificant in other outcomes	This lifestyle intervention for workers in the construction industry at risk for CVD had significant effects on snack and fruit intake and smoking at 6 months. The significant effects on snack intake were unchanged at the long term.	Limitations: (1) Over- and underreporting of behavior, may have attenuated the intervention effects. (2) No validated questionnaire was used for measuring dietary intake. (3) Participants were not blinded. (4) Selective participation and dropout. (5) By analyzing the EB and SC group separately, the study may have become underpowered. Strengths: (1) Compliance to the intervention was rather high. (2) Contrast between groups was large. (3) Randomization performed at individual level. (4) The participants in the EB group were analyzed separately from those in the SC group, in order to determine changes in the lifestyle behaviors that were

				actually aimed at. (5) The study can be considered as an effectiveness study as opposed to an efficacy study.
Brandt [18]	Physical workload	Number of events: No group-by-time interaction effect (P=.75) and (P=.51) General fatigue: within-group decrease both at 3-month and 6-month follow-up (P=.001) Own work: within-group increase at 3-month follow-up (P=0.04)	This PE intervention with technical measurements did not reduce the number of events with excessive physical workload during construction work. However, the intervention led to decreased general fatigue and increased influence on own work.	Limitations: (1) The number of dropouts was higher than expected, resulting in reduced statistical power. However, no participants drop out of the study due to a lack of willingness to participate but were missing at random, and all participant were included in the intention-to-treat analysis. (2) The size of the individual clusters was larger than anticipated, which also reduced the statistical power due to intra-cluster correlation. (3) Loss of data from sEMG, especially from the erector spinae muscles at baseline. (4) Inherent variation in daily working tasks at the construction sites, used steps and heart rate for controlling. (5) Only included large-

				scale construction companies. (6) The difference in age and, partly, experience between the intervention and control group, used age for controlling. Strengths: (1) Cluster RCT study design. (2) The use of technical measurements to quantify the workload rather than relying on self-reports or observations.
Groeneveld [19]	Lifestyle intervention for workers at risk for cardiovascular disease	A statistically significant effect of the intervention on body weight (β = -2.0, 95% CI: (-2.9, -1.2)).	The 6-month intervention had a significantly positive effect on body weight at 12-months follow-up. The intervention was more effective, but also more expensive than usual care.	Limitations: (1) Noncompletion was relatively large and partly selective. (2) The economic evaluation was underpowered to detect differences in costs, as the sample size calculation was based on a clinical outcome. (3) The lack of data on presenteeism. Strengths: (1) A full economic evaluation was performed within a study with RCT. (2) The cost- effectiveness of the intervention was not assessed until 6 months after the intervention had ended. (3) Sensitivity analyses were done to indicate the impact of a change in assumptions.

Oude Hengel [20]	Health and work ability (reduction of physical workload)	Mean sickness absenteeism costs per worker were significantly lower in the intervention group (-760; 95% CI: -1497; -156). Pooled effects: Work ability: -0.12 (-0.79, 0.54) Physical health: -0.80 (-3.22, 1.61) Mental health: 1.00 (-1.15, 3.15) Musculoskeletal symptoms: Back: -0.03 (-0.13, 0.07) Neck /shoulder: -0.05 (-0.14, 0.05) Upper extremities: -0.05 (-0.15, 0.04) Lower extremities: -0.01 (-0.13, 0.10)	The intervention in the present study was cost saving to the employer due to reduced sickness absenteeism costs in the intervention group compared with the control group. However, the intervention cannot be regarded as cost-effective as no significant effects were found for work ability and health.	Limitations: (1) The best way to conduct and interpret economic evaluation is still being debated. (2) High loss to follow-up on the outcomes. (3) no universal agreement about the most appropriate method for measuring or monetizing presenteeism costs. Strengths: (1) Randomization was performed at the department level. (2) Cluster RCT study design. (3) Sickness absence data were retrieved from continuous sickness absence registration systems of the six participating companies, which eliminated information or recall bias, and limited loss-to-follow up.
Gram [21]	Aerobic capacity and muscle strength	VO2max: IG: +0.3 (0.4) CG: -0.0 (0.4) Diff 0.4 (0.2, 0.5)* HR: IG: -10 (11) CG: 0 (10)	Training for 20 minutes, 3 times a week significantly increased VO2max with a clinically relevant magnitude regarding risk of	Limitations: Low acceptance within this job group to participate in exercise training.

		Diff -10 (-15.2,-4.6)* Relative VO2max: IG: 3.9 (4.4) CG: 0.3 (4.5) Diff 3.7 (1.5,5.9)* Blood pressure, cholesterol, isomeric muscle strength, BP, BMI are insignificant.	cardiometabolic disorders. This study demonstrates a good effectiveness for integrating short exercise bouts into organizational routines among constructions workers.	
Milner [22]	Mental health stigma	Adjusted mean difference: Help-seeking inhibition: -0.03 (-0.74, 0.68) Shame: -0.19 (-0.96, 0.58) Self-blame: -0.06 (-0.50, 0.39)	Self-stigma was relatively low in the sample. The intervention had no significant effect on self-stigma, after adjusting for confounders. There were reductions in stigma in both the intervention and control groups at 6-week follow-up. Process evaluation indicated that	Limitations: (1) Considerable drop out in the present study. (2) It would have been useful to have data on a participants' mental health and help-seeking behaviors following participation in this trial, in order to assess some of the correlates of stigma. (3) Lack of data on other possible confounders, such as education.

			participants generally enjoyed the program and felt that it was beneficial to their mental health.	
Viester [23]	Musculoskeletal symptoms, physical functioning, work ability, work-related vitality, work performance, and sickness absence	Work-related vitality: 6 months: 0.19 (-0.02, 0.40) 12 months: 0.01 (-0.22, 0.23) Work performance: 6 months: 0.13 (-0.13, 0.38) 12 months: -0.08 (-0.45, 0.28) Work ability: 6 months: 0.72 (-0.33, 1.77) 12 months: 0.53 (-0.59, 1.65) Physical functioning: 6 months: -0.29 (-6.38, 5.79) 12 months: 0.45 (-6.21, 7.10) Sickness absence (OR): 6 months: 0.86 (0.47, 1.58) 12 months: 1.19 (0.66, 2.15)	The results of this RCT did not show effects of the programme on musculoskeletal symptoms, physical functioning, work-related vitality, work performance, work ability, or sickness absence.	Limitations: (1) Power calculation was performed on the primary outcome measure of the study, i.e. body weight, so under power might happen for other outcomes. (2) Missing data on items of the work ability index resulted in a reduced number of complete cases. (3) All outcomes are obtained via self-report except absence record. (4) Behavior change in colleagues working at the same worksites could have influenced control participants inducing contamination. Strengths: (1) RCT study design. (2) Obtaining sickness absence data from company records.

All 4 musculoskeletal symptoms

also insignificance

Gram [24]	Aerobic capacity and muscle strength	Work ability: IG: 0.4 (1.6) CG: -0.1 (1.3) Diff 0.4 (-0.2, 1.1) Perceived exertion: IG: 0.1 (2.2) CG: -0.3 (1.6) Diff 0.1 (-0.6, 0.8) Productivity: IG: -0.3 (2.1) CG: -0.1 (1.1) Diff -0.5 (-1.4, 0.4) Sick leave: IG: 0.7 (2.2) CG: 0.1 (4.6) Diff -0.1 (-1.7, 1.4) All 8 measures of pain intensity are also insignificant.	Training for 20 minutes, 3 times a week significantly increased VO2max with a clinically relevant magnitude regarding risk of cardiometabolic disorders. This study demonstrates a good effectiveness for integrating short exercise bouts into organizational routines among constructions workers.	Limitations: The sample size calculation in this study was based on minimal difference in physical capacity after the intervention, study might be underpowered. A possible type 2 error concerning this variable cannot be excluded in this study. Strengths: RCT study design.
Oude Hengel [26]	Work ability, health, and sick leave	Overall effect Work ability: β = 0.02 (-0.34, 0.37) Physical health status: β = -0.04 (-1.43, 1.35) Mental health status:	The prevention program seemed to result in a beneficial but not statistically significant decline in the prevalence of	Limitations: (1) Most data were obtained from questionnaires collected at the worksite. (2) Limited statistical power. Power calculation was based on the number of cases needed to identify an effect on mental and physical health status and not the other

		β = 0.80 (-0.51, 2.11) Musculoskeletal symptoms (OR): from 0.68 (0.34, 1.33) to 0.86 (0.47, 1.57) Long-term sick leave (OR): 0.44 (0.13, 1.26)	musculoskeletal symptoms and long-term sick leave among construction workers, but showed no effects with regard to work ability, physical health, and mental health.	outcome measures. And the loss-to-follow-up was higher than expected due to the economic crisis. (3) A relative high rate of data was missing for the physical and mental health score. Strengths: (1) Cluster RCT study design and randomization at department and company level. (2) Current study population consisted of construction workers with different professions, from all over the Netherlands and of all ages. (3) Sick leave data were gathered from the continuous registration systems, which eliminated information or recall bias and limited loss-to-follow up.
Oude Hengel [27]	Work engagement, social support, physical workload, and need for recovery	Overall effect Overall social support: β = 0.03 (-0.39, 0.46) Work engagement: β = 0.02 (-0.12, 0.15) Physical workload: β = 0.10 (-0.02, 0.21)	The intervention neither improved social support nor work engagement, nor was it effective in reducing the physical workload and need for recovery among construction workers.	Limitations: (1) The study design was two-armed (control versus intervention), which does not allow a separate evaluation of the individual components of the prevention program. (2) Sample size calculation was based on a change in health status, which may cause underpower. (3) Data were obtained solely from questionnaires. (4) Participation in the program was voluntary, and bias due to non-response could therefore not be ruled out

				in intervention studies. (5) Loss-to-follow-up was higher than expected due to the economic crisis and health-related absenteeism of the workers. Strengths: (1) Cluster RCT study design. (2) High participation of construction workers. (3) Randomization at department level that minimized contamination.
de Boer [28]	Work ability and work disability pension	Work ability (WAI score): no overall improvement (p=0.52) no differences in improvement over time between the groups (p=0.09) Age-adjusted disability pension (diff in %): 9 months: 0.8 (-4.8, 6.5) 18 months: 5.4 (-3.5, 14.4) 26 months: 11.8 (0.0, 23.7)	The programme was slightly effective in improving the work ability but not in reducing work disability pensions. A more comprehensive multidisciplinary intervention programme might be necessary.	Limitations: (1) Selection of participants was found to be difficult. Fewer employees than expected were selected by their occupational physician via the selection instrument during the periodic occupational health examination. (2) The majority of participants enrolled themselves in the programme. This self-enrolment shows a great need for our training programme from building industry employees, but it resulted in fewer baseline data. (3) A considerable number of participants did not complete the programme. (57%)

Hammer [29]	Organizational context	Family Supportive Supervisor Behaviors (FSSB): B = -0.08, p=0.63 Team effectiveness processes: B = 0.11, p=0.34 Work-life effectiveness indicators: B=0.08, p=0.52	Main effects for the intervention on perceptions of family supportive supervisor behaviors, team effectiveness, and work-life effectiveness were not found. But found that the intervention was more beneficial for work crew members who had poorer pre-intervention perceptions of their supervisor (lower leader-member exchange) and lower perceived team cohesion.	Limitations: (1) Main effects not significant, and the moderation effects had small but significant effect sizes. (2) A systematic evaluation of the environment-intervention fit did not take place. Intervention demands might have created more stress than alleviated stress. (3) Self-reported data.

Table 4 Risk of bias in included RCTs (n = 19)

Study	Randomization type	Adequate sequence generation	Allocation concealment	Blinding of participants to allocation	Blinding of outcome	Incomplete outcome data addressed	Free of selective reporting	Free of other bias	Final evaluation
Sparer [13]	Cluster RCT	+	-	-	-	+	+	-	-
Seixas [25]	Cluster RCT and RCT	+	-	-	-	+	+	-	-
Deurssen [31]	Cluster RCT	+	-	-	+	+	+	-	+
Sorensen [33]	RCT	+	-	-	-	+	+	+	-
Peters [35]	Cluster RCT	+	-	-	-	+	+	-	-
Ludewig [36]	RCT	+	-	-	-	+	+	+	-
Viester [15]	RCT	+	-	-	+	+	?	+	+
Groeneveld [16]	RCT	+	-	-	+	+	+	+	+
Groeneveld [17]	RCT	+	-	-	-	+	+	+	-
Brandt [18]	Cluster RCT	+	+	-	+	+	+	+	+
Groeneveld [19]	RCT	+	+	-	+	+	+	-	+

Oude Hengel [20]	Cluster RCT	+	+	-	-	+	+	+	+
Gram [21]	RCT	+	-	-	+	+	?	+	+
Milner [22]	RCT	+	+	+	-	+	+	+	+
Viester [23]	RCT	+	+	-	+	+	?	-	-
Gram [24]	RCT	+	-	-	-	+	?	+	-
Oude Hengel [26]	Cluster RCT	+	+	-	-	?	?	+	+
Oude Hengel [27]	Cluster RCT	+	+	-	-	?	?	+	+
Hammer [29]	Cluster RCT	+	-	-	-	+	+	-	-

Table 5 Risk of bias in included non-randomized studies (n = 5)

Study	Selection of participants	Confounding variables	Measurement of intervention (exposure)	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Final evaluation
Marín [14]	-	-	+	-	+	+	-
Houdmont [30]	-	?	+	-	+	+	+
Dian [32]	-	?	+	-	+	+	-
Borstad [33]	-	-	-	-	+	+	-
de Boer [28]	-	?	+	-	+	+	-

References

[1] "Facts on safety at work," 2005.

[2] W. Yi and A. Chan, "Health Profile of Construction Workers in Hong Kong.," *Int. J. Environ. Res. Public Health*, vol. 13, no. 12, 2016.

[3] "Overview Report of Vital Statistics in FY 2015: Occupational and Industrial Aspects." [Online]. Available: https://www.mhlw.go.jp/english/database/db-hw/orvf/SummaryReport2015.html. [Accessed: 09-Dec-2019].

[4] V. Arndt, D. Rothenbacher, U. Daniel, B. Zschenderlein, S. Schuberth, and H. Brenner, "All-cause and cause specific mortality in a cohort of 20 000 construction workers; results from a 10 year follow up," *Occup. Environ. Med.*, vol. 61, no. 5, pp. 419–425, May 2004.

[5] G. Sorensen *et al.*, "Integrating worksite health protection and health promotion: A conceptual model for intervention and research," *Prev. Med. (Baltim).*, vol. 91, pp. 188–196, Oct. 2016.

[6] D. C. Ham, T. Przybeck, J. R. Strickland, D. A. Luke, L. J. Bierut, and B. A. Evanoff, "Occupation and workplace policies predict smoking behaviors: Analysis of national data from the current population survey," *J. Occup. Environ. Med.*, vol. 53, no. 11, pp. 1337–1345, 2011.

[7] N. Darmon and A. Drewnowski, "Contribution of food prices and diet cost to socioeconomic disparities in diet quality and health: a systematic review and analysis," *academic.oup.com*.

[8] G. Sorensen, E. B.-T. N. I. of O. S. and, and undefined 2004, "Steps to a healthier US workforce: integrating occupational health and safety and worksite health promotion: state of the science."

[9] "Research compendium: the NIOSH Total Worker Health (TM) Program: seminal research papers 2012.," May 2012.

[10] D. Moher *et al.*, "Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement," *PLoS Medicine*, vol. 6, no. 7. Jul-2009.

[11] J. P. Higgins and S. Green, *Cochrane Handbook for Systematic Reviews of Interventions: Cochrane Book Series*. 2008.

[12] S. Y. Kim *et al.*, "Testing a tool for assessing the risk of bias for nonrandomized studies showed moderate reliability and promising validity," *J. Clin. Epidemiol.*, vol. 66, no. 4, pp. 408–414, Apr. 2013.

[13] E. H. Sparer, P. J. Catalano, R. F. Herrick, and J. T. Dennerlein, "Improving safety climate through a communication and recognition program for construction: a mixed methods study.," *Scand. J. Work. Environ. Health*, vol. 42, no. 4, pp. 329–37, 2016.

[14] L. S. Marin and C. Roelofs, "Promoting Construction Supervisors' Safety-Efficacy to Improve Safety Climate: Training Intervention Trial," *J. Constr. Eng. Manag.*, vol. 143, no. 8, Aug. 2017.

[15] L. Viester, E. A. L. M. Verhagen, P. M. Bongers, and A. J. van der Beek, "Effectiveness of a Worksite Intervention for Male Construction Workers on Dietary and Physical Activity Behaviors, Body Mass Index, and Health Outcomes: Results of a Randomized Controlled Trial.," *Am. J. Health Promot.*, vol. 32, no. 3, pp. 795–

805, Mar. 2018.

[16] I. F. Groeneveld, K. I. Proper, A. J. van der Beek, and W. van Mechelen, "Sustained body weight reduction by an individual-based lifestyle intervention for workers in the construction industry at risk for cardiovascular disease: Results of a randomized controlled trial," *Prev. Med. (Baltim).*, vol. 51, no. 3–4, pp. 240–246, Sep. 2010.

[17] I. F. Groeneveld, K. I. Proper, A. J. Van Der Beek, V. H. Hildebrandt, and W. Van Mechelen, "Short and long term effects of a lifestyle intervention for construction workers at risk for cardiovascular disease: A randomized controlled trial," *BMC Public Health*, vol. 11, 2011.

[18] M. Brandt *et al.*, "Effects of a participatory ergonomics intervention with wearable technical measurements of physical workload in the construction industry: Cluster randomized controlled trial," *J. Med. Internet Res.*, vol. 20, no. 12, 2018.

[19] I. F. Groeneveld, M. F. Van Wier, K. I. Proper, J. E. Bosmans, W. Van Mechelen, and A. J. Van Der Beek, "Cost-effectiveness and cost-benefit of a lifestyle intervention for workers in the construction industry at risk for cardiovascular disease," *J. Occup. Environ. Med.*, vol. 53, no. 6, pp. 610–617, Jun. 2011.

[20] K. M. Oude Hengel, J. E. Bosmans, J. M. Van Dongen, P. M. Bongers, A. J. Van der Beek, and B. M. Blatter, "Prevention program at construction worksites aimed at improving health and work ability is cost-saving to the employer: Results from an RCT," *Am. J. Ind. Med.*, vol. 57, no. 1, pp. 56–68, Jan. 2014.

[21] B. Gram, A. Holtermann, K. Søgaard, and G. Sjøgaard, "Effect of individualized worksite exercise training on aerobic capacity and muscle strength among construction workers - a randomized controlled intervention study," *Scand. J. Work. Environ. Heal.*, vol. 38, no. 5, pp. 467–475, Sep. 2012.

[22] A. Milner, P. C. F. Law, C. Mann, T. Cooper, K. Witt, and A. D. LaMontagne, "A smart-phone intervention to address mental health stigma in the construction industry: A two-arm randomised controlled trial," *SSM-POPULATION Heal.*, vol. 4, pp. 164–168, Apr. 2018.

[23] L. Viester, E. A. L. M. Verhagen, P. M. Bongers, and A. J. van der Beek, "The effect of a health promotion intervention for construction workers on work-related outcomes: results from a randomized controlled trial," *Int. Arch. Occup. Environ. Health*, vol. 88, no. 6, pp. 789–798, Aug. 2015.

[24] B. Gram, A. Holtermann, U. Bültmann, G. Sjøgaard, and K. Søgaard, "Does an exercise intervention improving aerobic capacity among construction workers also improve musculoskeletal pain, work ability, productivity, perceived physical exertion, and sick leave?: A randomized controlled trial," *J. Occup. Environ. Med.*, vol. 54, no. 12, pp. 1520–1526, Dec. 2012.

[25] N. S. Seixas *et al.*, "A multi-component intervention to promote hearing protector use among construction workers.," *Int. J. Audiol.*, vol. 50 Suppl 1, pp. S46-56, Mar. 2011.

[26] K. M. O. Hengel, B. M. Blatter, H. F. van der Molen, P. M. Bongers, and A. J. van der Beek, "The effectiveness of a construction worksite prevention program on work ability, health, and sick leave: results from a cluster randomized controlled trial," *Scand. J. Work Environ. Heal.*, vol. 39, no. 5, pp. 456–467, Sep. 2013.

[27] K. M. O. Hengel *et al.*, "Effectiveness of an intervention at construction worksites on work engagement,

social support, physical workload, and need for recovery: results from a cluster randomized controlled trial," *BMC Public Health*, vol. 12, p. 1008, Nov. 2012.

[28] A. G. E. M. de Boer, A. Burdorf, C. van Duivenbooden, and M. H. W. Frings-Dresen, "The effect of individual counselling and education on work ability and disability pension: a prospective intervention study in the construction industry.," *Occup. Environ. Med.*, vol. 64, no. 12, pp. 792–7, Dec. 2007.

[29] L. B. Hammer, D. M. Truxillo, T. Bodner, A. C. Pytlovany, and A. Richman, "Exploration of the impact of organisational context on a workplace safety and health intervention," *Work Stress*, vol. 33, no. 2, SI, pp. 192–210, Apr. 2019.

[30] J. Houdmont, P. Madgwick, and R. Randall, "Sun safety in construction: A UK intervention study," *Occup. Med. (Chic. Ill).*, vol. 66, no. 1, pp. 20–26, Jan. 2016.

[31] E. Van Deurssen *et al.*, "Effectiveness of a Multidimensional Randomized Control Intervention to Reduce Quartz Exposure among Construction Workers," *Ann. Occup. Hyg.*, vol. 59, no. 8, pp. 959–971, Jan. 2015.

[32] D. He *et al.*, "Promoting contraceptive use more effectively among unmarried male migrants in construction sites in china: A pilot intervention trial," *Asia-Pacific Journal of Public Health*, vol. 24, no. 5. pp. 806–815, Sep-2012.

[33] J. D. Borstad *et al.*, "A longitudinal analysis of the effects of a preventive exercise programme on the factors that predict shoulder pain in construction apprentices," *Ergonomics*, vol. 52, no. 2, pp. 232–244, 2009.

[34] G. Sorensen *et al.*, "Tools for health: the efficacy of a tailored intervention targeted for construction laborers.," *Cancer Causes Control*, vol. 18, no. 1, pp. 51–59, Feb. 2007.

[35] S. E. Peters, M. P. Grant, J. Rodgers, J. Manjourides, C. A. Okechukwu, and J. T. Dennerlein, "A Cluster Randomized Controlled Trial of a Total Worker Health((R)) Intervention on Commercial Construction Sites.," *Int. J. Environ. Res. Public Health*, vol. 15, no. 11, Oct. 2018.

[36] P. M. Ludewig and J. D. Borstad, "Effects of a home exercise programme on shoulder pain and functional status in construction workers," *Occup. Environ. Med.*, vol. 60, no. 11, pp. 841–849, Nov. 2003.

[37] P. Sedgwick and N. Greenwood, "Understanding the hawthorne effect," *BMJ (Online)*, vol. 351. BMJ Publishing Group, 04-Sep-2015.

[38] P. Velentgas, N. A. Dreyer, and A. W. Wu, "Outcome Definition and Measurement," 2013.

[39] T. S. Kristensen, "Intervention studies in occupational epidemiology," *Occup. Environ. Med.*, vol. 62, no. 3, pp. 205–210, Mar. 2005.

[40] T. Asfar, K. L. Arheart, N. A. Dietz, A. J. Caban-Martinez, L. E. Fleming, and D. J. Lee, "Changes in cigarette smoking behavior among us young workers from 2005 to 2010: The role of occupation," *Nicotine Tob. Res.*, vol. 18, no. 6, pp. 1414–1423, Jun. 2016.

[41] A. Cavaliere, E. De Marchi, and A. Banterle, "Exploring the adherence to the mediterranean diet and its relationship with individual lifestyle: The role of healthy behaviors, pro-environmental behaviors, income, and education," *Nutrients*, vol. 10, no. 2, Feb. 2018.

[42] S. J. W. Robroek, F. J. van Lenthe, P. van Empelen, and A. Burdorf, "Determinants of participation in

worksite health promotion programmes: A systematic review," *International Journal of Behavioral Nutrition and Physical Activity*, vol. 6. 20-May-2009.

[43] J. S. Boschman, H. F. van der Molen, J. K. Sluiter, and M. H. W. Frings-Dresen, "Psychosocial work environment and mental health among construction workers," *Appl. Ergon.*, vol. 44, no. 5, pp. 748–755, 2013.